

# HelixTrack Database Schema

## HelixTrack Database Schema — 2026-06-21

### Overview

HelixTrack uses SQLite (development) or PostgreSQL (production). The shipped core schema spans **five Definition versions (V1-V5)** plus a **V5.6 migration** (Migration.V5.6.sql — “V5 → V6 Security Engine Enhancements”, dated 2025-10-19). The five Definition DDL files declare **106 distinct tables in total**:

| Version | DDL file           | Theme                                                                                               | Distinct CREATE TABLE |
|---------|--------------------|-----------------------------------------------------------------------------------------------------|-----------------------|
| V1      | Definition.V1.sql  | Core                                                                                                | 60                    |
| V2      | Definition.V2.sql  | JIRA Feature Parity<br>— Phase 1                                                                    | 12                    |
| V3      | Definition.V3.sql  | JIRA Feature Parity<br>— Complete<br>(Phases 2 & 3)                                                 | 17                    |
| V4      | Definition.V4.sql  | Parallel Editing<br>Support                                                                         | 6                     |
| V5      | Definition.V5.sql  | Documents<br>Extension<br>Integration & Cross-<br>Entity Linking<br>Security Engine<br>Enhancements | 11                    |
| —       | Migration.V5.6.sql | (creates<br>permission_cache,<br>security_audit;<br>ALTERs the audit<br>table)                      | (migration)           |

Source of truth: the DDL files under /Volumes/T7/Projects/helix\_track/core/Database/DDL/. Counts above are distinct CREATE TABLE [IF NOT EXISTS] names per file (each file’s tables are additive on top of the prior version).

The category breakdowns below were authored against an earlier V1-V3 snapshot and use illustrative per-section counts that do not re-derive the 106-table total above — re-derive any per-section count directly from the DDL when precision matters.

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## V1 Core Tables (57)

### Business Entities

| Table      | Purpose             | Status        |
|------------|---------------------|---------------|
| project    | Projects/workspaces | ✅ Implemented |
| ticket     | Issues/tasks        | ✅ Implemented |
| user       | Users               | ✅ Implemented |
| comment    | Comments            | ✅ Implemented |
| team       | Teams               | ✅ Implemented |
| attachment | File attachments    | ⚠️ Partial    |
| label      | Labels/tags         | ❌ Missing     |
| priority   | Priorities          | ❌ Missing     |
| resolution | Resolutions         | ❌ Missing     |
| component  | Components          | ❌ Missing     |
| cycle      | Sprints/iterations  | ❌ Missing     |

### Workflow Entities

| Table         | Purpose              | Status    |
|---------------|----------------------|-----------|
| workflow      | Workflow definitions | ❌ Missing |
| workflow_step | Workflow steps       | ❌ Missing |
| ticket_type   | Ticket types         | ❌ Missing |
| ticket_status | Ticket statuses      | ❌ Missing |

### Board Entities

| Table        | Purpose              | Status    |
|--------------|----------------------|-----------|
| board        | Kanban/Scrum boards  | ❌ Missing |
| board_column | Board columns        | ❌ Missing |
| board_ticket | Board-ticket mapping | ❌ Missing |

### User Entities

| Table     | Purpose    | Status    |
|-----------|------------|-----------|
| user_role | User roles | ❌ Missing |

| Table           | Purpose          | Status    |
|-----------------|------------------|-----------|
| permission      | Permissions      | ✗ Missing |
| user_preference | User preferences | ✗ Missing |

## System Entities

| Table        | Purpose         | Status    |
|--------------|-----------------|-----------|
| system_info  | System metadata | ✗ Missing |
| audit_log    | Audit trail     | ✗ Missing |
| notification | Notifications   | ✗ Missing |

## V2 Extension Tables (14)

### Enhanced Features

| Table               | Purpose              | Status    |
|---------------------|----------------------|-----------|
| ticket_relationship | Ticket relationships | ⚠ Partial |
| ticket_link         | Ticket links         | ⚠ Partial |
| ticket_watchers     | Ticket watchers      | ⚠ Partial |
| ticket_time_entry   | Time tracking        | ⚠ Partial |

### Workflow Extensions

| Table               | Purpose              | Status    |
|---------------------|----------------------|-----------|
| workflow_transition | Workflow transitions | ✗ Missing |
| workflow_condition  | Workflow conditions  | ✗ Missing |
| workflow_action     | Workflow actions     | ✗ Missing |

## V3 Advanced Tables

### Security

| Table            | Purpose            | Status        |
|------------------|--------------------|---------------|
| security_audit   | Security audit log | ✓ Implemented |
| permission_cache | Permission cache   | ✓ Implemented |
| account          | Account management | ✓ Implemented |

Note: `security_audit` and `permission_cache` are (re)created/expanded by the `Migration.V5.6.sql` Security-Engine migration, not only by an early version.

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## V4 Tables — Parallel Editing Support

Defined in `Definition.V4.sql` (6 distinct tables). Supports concurrent/parallel editing of entities. Re-derive the exact table list from the DDL file.

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## V5 Tables — Documents Extension & Cross-Entity Linking

Defined in `Definition.V5.sql` (11 distinct tables). Integrates the Documents extension and cross-entity linking. Re-derive the exact table list from the DDL file.

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## Table Relationships

### Core Relationships

project (1) — (N) ticket  
project (1) — (N) board  
project (1) — (N) cycle  
project (1) — (N) component  
ticket (1) — (N) comment  
ticket (1) — (N) attachment  
ticket (N) — (N) label  
user (1) — (N) ticket (assignee)  
user (1) — (N) comment  
team (N) — (N) user

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## Indexes

### Performance Indexes

- `idx_ticket_project_id` — Ticket lookup by project
  - `idx_ticket_status` — Ticket lookup by status
  - `idx_ticket_assignee` — Ticket lookup by assignee
  - `idx_comment_ticket_id` — Comment lookup by ticket
  - `idx_audit_created` — Audit log by date
  - `idx_audit_user_id` — Audit log by user
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# Migrations

## Migration History

- V1 — Initial core schema (Definition.V1.sql)
- V2 — JIRA feature parity, phase 1 (Definition.V2.sql)
- V3 — JIRA feature parity, complete / phases 2 & 3 (Definition.V3.sql)
- V4 — Parallel editing support (Definition.V4.sql)
- V5 — Documents extension & cross-entity linking (Definition.V5.sql)
- V5 → V6 — Security engine enhancements (Migration.V5.6.sql, 2025-10-19)

## Migration Files

- core/Database/DDL/Migration.V1.2.sql
  - core/Database/DDL/Migration.V2.3.sql
  - core/Database/DDL/Migration.V3.4.sql
  - core/Database/DDL/Migration.V5.6.sql (also mirrored at core/Application/Database/DDL/Migration.V5.6.sql)
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## Cross-references

- [Architecture](#)
- [API Reference](#)

**Sources verified 2026-06-22: /Volumes/T7/Projects/helix\_track/core/Database/DDL/Definition.V1.sql..Definition.V5.sql , /Volumes/T7/Projects/helix\_track/core/Database/DDL/Migration.V5.6.sql , /Volumes/T7/Projects/helix\_track/core/Application/Database/DDL/Migration.V5.6.sql**

Reconciled 2026-06-22: schema range **V1-V3 / “71 tables” → V1-V5 + V5.6 migration / 106 distinct tables**, per the actual Definition.V\*.sql DDL (authoritative source of truth). Per-version distinct CREATE TABLE counts read directly from the DDL: V1=60, V2=12, V3=17, V4=6, V5=11 (total 106); the Migration.V5.6.sql “V5 → V6 Security Engine Enhancements” (2025-10-19) creates permission\_cache + security\_audit and ALTERs the audit table.

REPO-STATE-DERIVED (per §11.4.99 the sources are the cross-referenced DDL files, following the docs/ARCHITECTURE.md precedent). Cross-referenced on 2026-06-22: - **Schema DDL CONFIRMED present:** helix\_track/core/Database/DDL/Definition.V1.sql ... Definition.V5.sql exist (plus Indexes\_Performance.sql, Migration.V\*.sql), and the V5.6 migration is mirrored at core/Application/Database/DDL/ (Migration.V5.6.sql,

Test\_Data\_Users\_Permissions.sql). Extension and service DDLs (Extensions/{Chats,Documents,Times}/, Services/{Authentication,Localization}/) also exist and are NOT counted in the 106-table core total above. - **Resolved (version-range drift):** the prior revision stated “71 tables across **V1-V3**”; the live DDL tree carries Definition files **V1 through V5** plus a **V5.6** migration, with 106 distinct core tables. The overview table + Migration History have been re-derived from the V1-V5 DDL. The legacy per-section category counts (V1 “57”, V2 “14”) are illustrative groupings authored against the old V1-V3 snapshot and are NOT the authoritative totals — the 106-table figure from the DDL is. Re-verify any per-section count directly from the Definition.V\*.sql DDL, which remains the single source of truth.